

ORM Leaking More Than You Joined For

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ORM Leaking More Than You Joined For - elttam

By

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ORM Leaking More Than You Joined For

Developers are still exposing ORMs through robust filtering and search functionality.

ORM

ORM Leaks

web

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Introduction

In this article, we expand on our previous Object Relational Mapper (ORM) Leak research and [Black Hat EU briefing](#) by showcasing an interesting expression parser bug in the [Beego ORM](#) that we used to bypass ORM Leak protections in [Harbor](#). We also demonstrate an authentication bypass technique for the Prisma ORM that was omitted from our original [Prisma article](#).

We further argue that the introduction of an ORM Leak vulnerability does not depend on the use of a susceptible ORM such as [Django](#) or [Prisma](#). Developers frequently implement robust filtering

capabilities that parse user input into filter expressions for the underlying ORM of their choice, yet it is commonly overlooked that filtering on sensitive attributes should be disallowed.

To demonstrate this, we show how Microsoft's OData API can unintentionally expose sensitive attributes and Entity Framework filtering functionality. This behaviour was abused during a project earlier this year, where we were also able to bypass naive protections intended to prevent filtering of sensitive data.

We have also published `semgrep` rules for detecting potentially dangerous uses of the Django, Prisma, Beego, and Entity Framework ORMs at <https://github.com/elttam/semgrep-rules>.

Recap of the Previous ORM Leak Articles

plORMbing your Django ORM

In this article, we introduced the ORM Leak vulnerability class and demonstrated vulnerable uses of the Django ORM that allowed filtering on sensitive fields. We showcased a variety of exploitation techniques, including many-to-many relational filtering payloads that pivot through relationships to reach sensitive fields on connected objects. We also highlighted that response length differences are not the only viable oracle for ORM Leak attacks, showing how ReDoS payloads on a MySQL database can be used to perform error-based attacks.

plORMbing your Prisma ORM with Time-based Attacks

This article demonstrated that the Prisma ORM is also susceptible to similar relational filtering attacks to those showcased in our Django article, when developers allow users to control the `where` option, albeit with a different filtering syntax. Since Prisma provides greater control over the generated SQL queries compared to Django, we explored time-based attacks, which led to the development of `plormber`, a tool for exploiting time-based ORM Leak vulnerabilities.

Beego ORM Leaks

`Beego` is one of the most popular Golang web frameworks (over 32k GitHub stars). Its filtering syntax is heavily inspired by the Django ORM, as shown in the example below.

```
o := orm.NewOrm()
search := c.Ctx.Input.Query('search')
qs := o.QueryTable('articles')
qs = qs.Filter('title__contains', search)
var articles []*Article
_, err := qs.All(&articles)
```

Similarly, the Beego ORM adopts the same convention as Django for relational filtering.

```
qs.Filter('created_by__user__username__contains', search)
```

As a result, an ORM Leak vulnerability can be easily introduced when using the Beego ORM if a developer allows a user to control the filter expression parameter for the `Filter` function.

```
qs.Filter(filter_expression, filter_value) <1>
```

<1> Vulnerable to ORM leaks if the user controls the `filter_expression` parameter.

All the ORM Leak attacks discussed in our [plORMbing your Django ORM blog article](#) also apply to the Beego ORM. For this reason, we initially chose not to publish our findings that Beego was similarly susceptible.

However, earlier this year we decided to see how quickly we could discover an ORM Leak vulnerability in a large open-source project, and we came across this code snippet in [Harbor](#) in under 20 minutes.

!

Harbor CVE-2025-30086 - Initial Vulnerability

[Harbor](#) is a popular open source container registry (over 27k GitHub stars) built using the Beego web framework, and it relies on the Beego ORM for managing application data. The snippet above was located in the `setFilters` function shown below. This function was used to parse the `q` URL parameter for all `List` API handler endpoints.

`setFilters` function in [goharbor/harbor/blob/v2.12.2/src/lib/orm/query.go](#)

```

// set filters according to the query
func setFilters(ctx context.Context, qs orm.QuerySeter, query *q.Query, meta *metadata) orm.QuerySeter {
    for key, value := range query.Keywords {
        // The 'strings.SplitN()' here is a workaround for the incorrect usage of query which should be avoided
        // e.g. use the query with the knowledge of underlying ORM implementation, the 'OrList' should be used ins
        // https://github.com/goharbor/harbor/blob/v2.2.0/src/controller/project/controller.go#L348
        k := strings.SplitN(key, orm.ExprSep, 2)[0]
        mk, filterable := meta.Filterable(k)
        if !filterable {
            // This is a workaround for the unsuitable usage of query, the keyword format for field and method sh
            // e.g. 'ArtifactDigest' or the snake case format 'artifact_digest' should be used instead:
            // https://github.com/goharbor/harbor/blob/v2.2.0/src/controller/blob/controller.go#L233
            mk, filterable = meta.Filterable(snakeCase(k))
            if !filterable {
                continue
            }
        }
        // filter function defined, use it directly
        if mk.FilterFunc != nil {
            qs = mk.FilterFunc(ctx, qs, key, value)
            continue
        }
        // fuzzy match
        if f, ok := value.(*q.FuzzyMatchValue); ok {
            qs = qs.Filter(key+'__icontains', Escape(f.Value))
            continue
        }
        // range
        if r, ok := value.(*q.Range); ok {
            if r.Min != nil {
                qs = qs.Filter(key+'__gte', r.Min)
            }
            if r.Max != nil {
                qs = qs.Filter(key+'__lte', r.Max)
            }
            continue
        }
        // or list
        if ol, ok := value.(*q.OrList); ok {
            if ol == nil || len(ol.Values) == 0 {
                qs = qs.Filter(key+'__in', nil)
            } else {
                qs = qs.Filter(key+'__in', ol.Values...)
            }
        }
    }
}

```

```

        continue
    }
    // and list
    if _, ok := value.(*q.AndList); ok {
        // do nothing as and list needs to be handled by the logic of DAO
        continue
    }
    // exact match
    qs = qs.Filter(key, value) <1>
}
return qs
}

```

<1> The ORM Leak sink that was identified by searching for the `qs.Filter\([a-zA-Z0-9_]*, \s*[a-zA-Z0-9_]*\)` pattern on Sourcegraph.

For example, the `q` URL parameter for `GET /api/v2.0/users?q=email=~elttam.com` would be parsed as the `email_icontains` filter expression for the Beego ORM, returning all users whose email addresses contain `elttam.com`. However, there were no protections to prevent users from filtering on sensitive attributes such as `password` or `salt`.

!

Oddly, Harbor did not use the relational features of the ORM and instead relied on custom SQL for applying table joins in specific cases. This inadvertently reduced the exploitability of the ORM Leak vulnerability, as we could not use relational filtering techniques on endpoints accessible to low privileged or unauthenticated users. Exploitation required sufficient permissions to list other users.

This basic ORM Leak pattern is a common mistake we observe during engagements: an endpoint parses a user controllable filter constraint for querying users without preventing filtering on sensitive attributes.

What made this specific vulnerability in Harbor particularly interesting—and worth publishing—was our ability to repeatedly bypass Harbor's patches due to how the Beego ORM parses filter expressions in its `Filter` method.

Harbor CVE-2025-30086 - Bypassing the Deny List Patch

In Harbor version `v2.13.0`, the `parseFilterable` validation function was introduced to set the `Filterable` attribute of a model field to `true` if the field does not have the `filter` annotation set to `false` when [parsing a data model](#), and to `false` otherwise. Sensitive model fields were then annotated with `filter:'false'`, as shown in the following snippet from the `User` data model.

Snippet of the `User` model in `goharbor/harbor/blob/v2.13.0/src/pkg/user/dao/user.go`, where the `Password` and `Salt` fields are marked as not filterable.

```

type User struct {
    ...
    Password string `orm:'column(password)' filter:'false' json:'password'`
    ...
    Salt    string `orm:'column(salt)' filter:'false' json:'-'`
    ...
}

```

The `Filterable` attribute was then validated when parsing the user-controlled `q` parameter in `setFilters` (shown below). If it was `false`, the corresponding filter constraint was removed.

`setFilters` function in [goharbor/harbor/blob/v2.13.0/src/lib/orm/query.go](https://github.com/goharbor/harbor/blob/v2.13.0/src/lib/orm/query.go)

```

// set filters according to the query
func setFilters(ctx context.Context, qs orm.QuerySeter, query *q.Query, meta *metadata) orm.QuerySeter {
    for key, value := range query.Keywords {
        // The 'strings.SplitN()' here is a workaround for the incorrect usage of query which should be avoided
        // e.g. use the query with the knowledge of underlying ORM implementation, the 'OrList' should be used instead
        // https://github.com/goharbor/harbor/blob/v2.2.0/src/controller/project/controller.go#L348
        k := strings.SplitN(key, orm.ExprSep, 2)[0] <1>
        mk, filterable := meta.Filterable(k) <2>
        if !filterable {
            // This is a workaround for the unsuitable usage of query, the keyword format for field and method sh
            // e.g. 'ArtifactDigest' or the snake case format 'artifact_digest' should be used instead:
            // https://github.com/goharbor/harbor/blob/v2.2.0/src/controller/blob/controller.go#L233
            mk, filterable = meta.Filterable(snakeCase(k))
            if mk == nil || !filterable {
                continue
            }
        }
        ...
        // exact match
        qs = qs.Filter(key, value) <3>
    }
    return qs
}

```

<1> Splits the user controllable filter expression and retrieves the first section as the *filter key*.

<2> Validates if the *filter key* was filterable.

<3> ORM Leak sink.

Note that Harbor only validated the **start of the filter expression**, so if Harbor did utilise the relational features of the Beego ORM, this protection could have been easily bypassed using

relational filters. However, we were not afforded that luxury.

We did, however, notice an interesting quirk in how the Beego ORM parses non-relational fields within a filter expression via the `parseExprs` function. The filter expression is first split by the `_` expression section separator, and the `parseExprs` function then iterates over each segment in a loop. If a non-relational field is used as though it were a relational field—such as `email` in the `email_password` filter expression—it is **overwritten by the subsequent field in the expression**.

In practice, this means the filter expression `email_password_startswith` is equivalent to `password_startswith` when using the Beego ORM, which conveniently bypasses Harbor's deny-list patch.

!

Harbor CVE-2025-30086 - Bypassing Filter Expression Limitations

The second patch Harbor implemented in `v2.13.1-rc1` restricted the filter expression in the `q` input to allow only a single use of the Beego expression separator (`_`). The intention was to still allow users to choose a Beego operator for filtering fields, while preventing the `email_password_startswith` bypass described in the previous section.

[goharbor/harbor/blob/v2.13.1-rc1/src/lib/orm/query.go](https://github.com/goharbor/harbor/blob/v2.13.1-rc1/src/lib/orm/query.go)

```
// set filters according to the query
func setFilters(ctx context.Context, qs orm.QuerySetter, query *q.Query, meta *metadata) orm.QuerySetter {
    for key, value := range query.Keywords {
        // The 'strings.SplitN()' here is a workaround for the incorrect usage of query which should be avoided
        // e.g. use the query with the knowledge of underlying ORM implementation, the 'OrList' should be used instead
        // https://github.com/goharbor/harbor/blob/v2.2.0/src/controller/project/controller.go#L348
        keyPieces := strings.Split(key, orm.ExprSep)
        if len(keyPieces) > 2 { <1>
            log.Warningf("The separator '%s' is not valid in the query parameter '%s__%s'. Please use the correct field name.", orm.ExprSep, keyPieces[0], keyPieces[1])
            continue
        }
        ...
    }
}
```

<1> Only allows one occurrence `_` in the filter expression.

The issue with this patch is that Harbor maintains its own mapping to Beego operators based on the format of the `value`.

The `Build` function in [goharbor/harbor/blob/v2.13.1-rc1/src/lib/q/builder.go](https://github.com/goharbor/harbor/blob/v2.13.1-rc1/src/lib/q/builder.go)

```

// Build query sting, sort and pagination information into the Query model
// query string format: q=k=v,k=~v,k=[min~max],k={v1 v2 v3},k=(v1 v2 v3)
// exact match: k=v
// fuzzy match: k=~v <1>
// range: k=[min~max]
// or list: k={v1 v2 v3}
// and list: k=(v1 v2 v3)
// sort format: sort=k1,-k2
func Build(q, sort string, pageNumber, pageSize int64) (*Query, error) {
    keywords, err := parseKeywords(q)
    ...
    return &Query{
        Keywords: keywords,
        ...
    }, nil
}

```

<1> The *fuzzy match* Harbor query format.

Fuzzy match mapping to `__icontains` operator in `goharbor/harbor/blob/v2.13.1-rc1/src/lib/orm/query.go`

```

func setFilters(ctx context.Context, qs orm.QuerySetter, query *q.Query, meta *metadata) orm.QuerySetter {
    for key, value := range query.Keywords {
        ...
        // fuzzy match
        if f, ok := value.(*q.FuzzyMatchValue); ok {
            qs = qs.Filter(key+'__icontains', Escape(f.Value)) <1>
            continue
        }
        ...
    }
}

```

<1> Inserts the insensitive contains (`__icontains`) operator if the query value is a *fuzzy match*.

Using this, Harbor's second patch was bypassed by leveraging the *fuzzy match* value format with the `email__password` filter expression. However, the case sensitivity of the `salt` field was lost due to the use of the `__icontains` Beego operator.

!

Harbor CVE-2025-30086 - The Final Patch

In Harbor version `v2.14.0`, the ORM Leak vulnerability was finally patched by validating the second section of a filter expression against a set of allowed Beego operators. This effectively validates the entire filter expression input and prevents payloads such as `email__password=~`.

The operator allow list check within `setFilters` in `goharbor/harbor/blob/v2.14.0/src/lib/orm/query.go`


```
// only accept the below operators
if len(keyPieces) == 2 {
    operator := orm.ExprSep + keyPieces[1]
    allowedOperators := map[string]struct{}{
        '_icontains': {},
        '_in':        {},
        '_gte':       {},
        '_lte':       {},
        '_gt':        {},
        '_lt':        {},
        '_exact':    {},
    }
    if _, ok := allowedOperators[operator]; !ok {
        log.Warningf('the operator '%s' in query parameter '%s' is not supported, the query will be skipped.', operator, key)
        continue
    }
}
```

Key Takeaways from Harbor CVE-2025-30086

There are three key takeaways from the ORM Leak vulnerability we discovered in Harbor:

Robust Filtering Capabilities

One root cause of this vulnerability was that the Harbor API allowed user inputs to set both the **field** and **operator** for filtering users, without validating against **sensitive fields**.

This type of issue is **very common** in our engagements, even on targets under bug bounty programs. Basic ORM Leak vulnerabilities appear more frequently than similar classes like SQL injection.

The Beego ORM Filter Expression Parsing is Broken

The broken parsing in the `parseExprs` function can be exploited to bypass naive ORM Leak protections that do not validate the entire filter expression containing user-controlled sections.

We reported this parsing issue to the Beego contributors in May 2025, but did not receive an adequate response. The issue is still present in the latest version (v2.3.8) at the time of this article.

This is not the first filter expression parsing bug discovered that could bypass ORM Leak protections. For example, [Strapi - CVE-2023-34235](#) allowed insertion of a `WHERE` SQL condition with a field that did not exist on the corresponding model, bypassing Strapi's patch for [CVE-2023-22894](#).

Application Mappings to ORM Operators

Developers often implement mappings from user input to ORM operators, which can inadvertently introduce ORM Leak vulnerabilities.

During another engagement, we observed this with the [Sequelize ORM](#), which uses a syntax similar to [Prisma ORM](#). While Sequelize itself is generally resistant to ORM Leaks because its operators are JavaScript `Symbol` types, a middleware in the application we were testing mapped string inputs to Sequelize operators, resulting in an ORM Leak vulnerability.

Prisma ORM Authentication Bypass

In our original [Prisma ORM Leak article](#), we overlooked including a common mistake that we have observed: assuming the type of a user input when filtering by a field.

For example, the following TypeScript snippet shows a password reset endpoint in an [ExpressJS](#) application that resets the password for the first user where the `resetToken` field is *equal* to the provided `resetToken` input.

```
const app = express();
app.use(express.json())

app.post('/user/reset-password', async (req, res) => {
  if ((req.body.resetToken ?? "") === "") {
    res.status(400).json({ 'msg': 'invalid code' })
    return
  }
  try {
    const user = await prisma.user.findFirstOrThrow({
      where: {
        resetToken: req.body.resetToken as string <1>
      }
    })
    await prisma.user.update({
      where: {id: user.id},
      data: {password: req.body.password, resetToken: null}
    })
    res.json({ 'msg': `user ${user.email} had their password reset` })
  } catch (error) {
    console.error(error)
    res.status(400).json({ 'msg': 'invalid code' })
  }
})
```

<1> Assumption that the `resetToken` input is a `string` type.

Ignoring the insecure [non-constant time comparison](#) when comparing a sensitive value within a database query, there is an implicit assumption that the user input is of type `string` and that a simple equality comparison will be performed. However, as [mentioned previously](#), the Prisma ORM uses an object-based syntax, where operators are defined using **objects**.

In this case the `resetToken` value in the JSON request can be manipulated into an **object** that injects a Prisma operator. In the following example, this bypasses the reset token validation by injecting the **not** condition, which is conceptually similar to [NoSQL injection](#).

!

Another useful operator for bypassing authentication checks in Prisma is `contains`, particularly when you want to avoid compromising the first user in the users table.

!

JSON request bodies are not the only way to coerce user input into an object type in ExpressJS (one of the most common frameworks we see used with Prisma).

For URL-encoded request bodies, if the extended option is enabled for the middleware (`app.use(express.urlencoded({ extended: true }))`), an input such as `resetToken[not]=E` is parsed into the object `{'resetToken': {'not': 'E'}}`, injecting the **not** condition.

Similarly, [prior to ExpressJS version 5.x](#), URL parameters were parsed using the extended parser by default, allowing inputs like `resetToken[not]=E` to be coerced into objects. While [the default URL parameter parser was changed to the “simple” parser in ExpressJS 5.x](#), we still commonly observe the extended parser enabled in ExpressJS 5.x applications.

Another commonly overlooked input vector is cookies. If the [ExpressJS cookie-parser middleware is enabled](#), it supports *JSON Cookies*, which parse cookie values prefixed with `j:` using `JSON.parse`. A header such as `Cookie: resetToken=j:{'not': 'E'}` would result in the `resetToken` cookie being parsed into an object, injecting the **not** condition when used in a Prisma filter.

In summary, when input type validation is missing and the Prisma ORM is used, the resulting impact can be severe.

Leaking Non-Susceptible ORMs

As mentioned in the key takeaways section on the Beego ORM, one point we want to clarify for software engineers and security testers is that the introduction of an ORM Leak vulnerability **does not depend on using a susceptible ORM such as Django, Prisma, or Beego**.

The most common root cause we observe is the growing trend for web applications to offer *robust filtering or search capabilities*, which can be abused to filter objects by sensitive or hidden fields. Developers often rely on the ORM to determine which fields are queryable and to prevent SQL injection, but overlook explicitly validating which fields **should** be queryable.

A good example of this issue arises with *robust filtering* features in applications using the **Entity Framework ORM**, which we consider **not inherently susceptible to ORM Leaks**. Introducing an ORM Leak using Entity Framework typically requires additional custom parsing logic for a user input. By contrast, demonstrating ORM Leaks with Django, Beego, or Prisma often requires only a single line of code.

To illustrate the code required to introduce an ORM Leak vulnerability in Entity Framework, the following snippet is taken from [Microsoft's documentation on LINQ expressions and constructing runtime queries](#). This pattern would be vulnerable to ORM Leaks if the `source` Entity Framework model contained a sensitive field of type `string` and an attacker could control the search `term`.

Microsoft - Query based on run-time state

```

// using static System.Linq.Expressions.Expression;

IQueryable<T> TextFilter<T>(IQueryable<T> source, string term)
{
    if (string.IsNullOrEmpty(term)) { return source; }

    // T is a compile-time placeholder for the element type of the query.
    Type elementType = typeof(T);

    // Get all the string properties on this specific type.
    PropertyInfo[] stringProperties = elementType
        .GetProperties()
        .Where(x => x.PropertyType == typeof(string)) <1>
        .ToArray();
    if (!stringProperties.Any()) { return source; }

    // Get the right overload of String.Contains
    MethodInfo containsMethod = typeof(string).GetMethod('Contains', [typeof(string)]!);

    // Create a parameter for the expression tree:
    // the 'x' in 'x => x.PropertyName.Contains('term')'
    // The type of this parameter is the query's element type
    ParameterExpression prm = Parameter(elementType);

    // Map each property to an expression tree node
    IEnumerable<Expression> expressions = stringProperties
        .Select(prp =>
            // For each property, we have to construct an expression tree node like x.PropertyName.Contains('term')
            Call(
                // .Contains(...)
                Property(
                    // .PropertyName
                    prm,
                    // x
                    prp
                ),
                containsMethod,
                Constant(term) // 'term' <2>
            )
        );

    // Combine all the resultant expression nodes using ||
    Expression body = expressions
        .Aggregate((prev, current) => Or(prev, current));

    // Wrap the expression body in a compile-time-typed lambda expression
    Expression<Func<T, bool>> lambda = Lambda<Func<T, bool>>(body, prm);

```

```
// Because the lambda is compile-time-typed (albeit with a generic parameter), we can use it with the Where method
return source.Where(lambda);
}
```

<1> Includes all properties of the given `T` data model that are a `string` type, which could contain sensitive fields.

<2> Will also search by the value of sensitive fields.

Generated SQL that filtered sensitive `string` fields using the user controllable `term` input.

```
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
      Executed DbCommand (3ms) [Parameters=[@__p_0='?' (DbType = Int32), @__p_1='?' (DbType = Int32)], CommandType=Text
      SELECT [u].[Id], [u].[Username]
      FROM [Users] AS [u]
      WHERE [u].[Username] LIKE N'%attackerinput%' OR [u].[Password] LIKE N'%attackerinput%' OR [u].[ApiToken] LIKE N'%attackerinput%'
      ORDER BY [u].[Username]
      OFFSET @__p_0 ROWS FETCH NEXT @__p_1 ROWS ONLY
```

It is a common mistake in web applications for utility search functions to inadvertently filter on sensitive fields within a data model.

[Directus CVE-2025-64748, published last month](#), is a good example of this issue. In this case, the application included the `token` and `tfa_secret` fields when searching objects in the `directus_users` collection, as shown in the generated SQL from Directus.

```
select `directus_users`.`token`, `directus_users`.`id` from `directus_users` where (
  ...
  LOWER(`directus_users`.`tfa_secret`) LIKE ? or
  ...
  LOWER(`directus_users`.`token`) LIKE ? or
  ...
) order by `directus_users`.`id` asc limit ?
```

There are also libraries that add advanced filtering capabilities to API endpoints by integrating with an underlying ORM, which can inadvertently introduce ORM Leak vulnerabilities.

[Ransacking your password reset tokens](#) is an excellent article by Lukas Euler describing how the [Ransack library for ActiveRecord](#) allowed filtering on all model attributes by default.

Another library we encountered during an engagement earlier this year that did not enforce an allow list for queryable fields was [Microsoft's ASP.NET OData Web API](#).

Leaking EntityFramework using OData Query Endpoints

OData is a protocol for RESTful APIs that was first released in 2007 and standardises CRUD operations and **filter syntax**.

Security issues affecting OData APIs have been well known for over a decade, which is why we want to focus specifically, from a development perspective, on how sensitive fields can be inadvertently exposed in an ASP.NET OData API. We also discuss techniques we used during an engagement this year to bypass protections that did not follow [best practices](#).

The [OData standard defines the Entity Data Model \(EDM\)](#) as an abstract metadata model that describes entity types, their properties, and their associations with other entity types exposed by OData API endpoints.

For example, in the following `Article` and `User` models, the `Article` model has an association to a `User` object via the `CreatedBy` field.

```
public class Article
{
    public int Id { get; set; }
    public string Title { get; set; }
    public string Body { get; set; }
    public User CreatedBy { get; set; }
}
```

```
public class User
{
    public int Id { get; set; }
    public required string Username { get; set; }
    public required string Password { get; set; }
    public string? ApiToken { get; set; } // hex string
    public bool IsAdmin { get; set; }
}
```

Adding data models to the EDM is done using the `odataconventionmodelbuilder` class. In the following snippet, it is important to note that **only the `Article` model** was explicitly included.

```
var modelBuilder = new ODataConventionModelBuilder();
modelBuilder.EntitySet<Article>('Articles');
```

The EDM can then be retrieved from the `/metadata` endpoint. In our example application, the **`User` model was automatically included in the EDM due to its association with the `Article` model, even though it was not explicitly added.**

!

This is a common issue with OData APIs, where sensitive fields can be inadvertently included in the EDM and potentially exposed via an OData query endpoint using the `$expand` option.

OData controller for the `Articles` that has an OData query endpoint

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.AspNetCore.OData.Query;
using Microsoft.AspNetCore.OData.Routing.Controllers;
using ODataLeakDemo.Models;

namespace ODataLeakDemo.Controllers
{
    public class ArticlesController : ODataController
    {

        private readonly DataContext ctx;

        public ArticlesController(DataContext ctx)
        {
            this.ctx = ctx;
        }

        [EnableQuery] <1>
        public ActionResult<IQueryable<Article>> Get()
        {
            return Ok(ctx.Articles); <2>
        }
    }
}
```

<1> Enables OData query options.

<2> The OData API parses user inputs into a LINQ expression that is then applied to the returned `IQueryable<T>` , which in this case are the `Article` objects from EntityFramework.

Using the `$expand` option to leak the `CreatedBy` association.

!

The [best practice for sensitive fields is to exclude the field from the EDM using the `IgnoreDataMember` annotation](#). However, this is a deny list approach that could result in the developer not adding the `IgnoreDataMember` annotation for a sensitive field or the developer not being aware of the `IgnoreDataMember` annotation, which is what we observed during an engagement.

In that engagement, the developer used the following protections to mitigate against the leaking of sensitive fields:

- All OData options were disabled, except for the `$filter` option.
- Sensitive fields were marked using a custom annotation that disabled the use of all OData functions (e.g., `startswith` and `contains`) to prevent search operations on those fields.

To replicate these protections, we just added the `AllowedQueryOptions = AllowedQueryOptions.Filter` and `AllowedFunctions = AllowedFunctions.None` options to our OData query endpoint in a demo application, as shown below:

```
[EnableQuery(
    AllowedQueryOptions = AllowedQueryOptions.Filter,
    AllowedFunctions = AllowedFunctions.None
)]
public ActionResult<IQueryable<Article>> Get()
{
    return Ok(ctx.Articles);
}
```

The first protection was easily bypassed, as we could still write OData filter expressions on sensitive fields as part of an ORM Leak attack. However, the `startswith` and `contains` functions remained disabled.

To work around this, we used [logical operators](#) that were not disabled to perform comparisons based on the **database collation** (discussed in the next section) between a sensitive field and a string we controlled, allowing us to leak the full value.

!

Key Takeaways

Logical Operators Are Commonly Overlooked as a Search Mechanism

Logical operators such as greater than (`>`, `gt`, etc.) and less than (`<`, `lt`, etc.) can be used in ORM Leak attacks, yet are often overlooked by developers as a means of searching sensitive field values. We have used logical operators on multiple occasions to successfully exploit ORM Leak vulnerabilities.

Any ORM Could Be Leaked

With Entity Framework, we have shown that introducing an ORM Leak vulnerability typically requires more code than with more susceptible ORMs. However, this does not prevent developers from implementing filtering input parsers that can be abused to query the values of sensitive fields.

A Note About Database Collation

One concept we have omitted from our ORM Leak articles for the sake of simplicity is **database collation**, which defines the rules a database uses to store and **compare** data. Database collation can significantly impact the exploitation of ORM Leak vulnerabilities. For example, losing case-sensitivity due to the use of a case-insensitive collation, where the following databases use a case-insensitive collation by default:

- MariaDB

- MySQL
- SQLite
- MSSQL

Additionally, some ORMs attempt to handle case sensitivity at the application layer rather than relying on the database collation (e.g., Django and Beego), while others defer entirely to the database collation (e.g., Prisma and Entity Framework). Even then, ORMs that attempt to enforce case sensitivity may fail for certain database backends.

[Louis Nyffenegger from PentesterLab](#) provides a good example of this with Django and SQLite. In this case, the Django ORM loses case sensitivity for SQLite databases when using the `__startswith` or similar operators because the `LIKE` SQL condition in `sqlite3` is case-insensitive. In contrast, the `__regex` operator preserves case sensitivity.

Another complication is that byte ordering in the database may not align with raw byte values due to collation rules, which are primarily designed to support different languages and character sets with varying sort orders. For example, using the default `SQL_Latin1_General_CP1_CI_AS` collation in MSSQL, sorting a string produces results that are not ordered by byte value (as shown below). This directly affects logical operator based ORM Leak attacks, which rely on predictable ordering.

String of characters that were sorted by ascending byte order using `latin1` encoding.

```
!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
```

Getting the actual character ordering of the default `SQL_Latin1_General_CP1_CI_AS` collation

```
1> DECLARE @input_string NVARCHAR(MAX) = ' !#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~'
2>
3> WITH SortedChars AS (
4>   SELECT SUBSTRING(@input_string, number, 1) AS Char
5>   FROM master.dbo.spt_values
6>   WHERE type = 'P' AND number <= LEN(@input_string)
7> )
8> SELECT STRING_AGG(Char, '') WITHIN GROUP (ORDER BY Char COLLATE DATABASE_DEFAULT) AS SortedString
9> FROM SortedChars;
10> GO
SortedString

' !#$%&'()*+,-./;?@[^\^_`{|}~+<=>0123456789AabBCcdDEefFGghHIijKklLMmnNOopPQqrRSstTUuvVWwxXYyz <1>

(1 row affected)
```

<1> Symbols are treated as special characters that have a lower sort value than alphanumeric characters when using the `SQL_Latin1_General_CP1_CI_AS` collation.

Using the `SQL_Latin1_General_CP1_CI_AS` sorted string in a logical operator ORM Leaks attack.

!

Below is a list of SQL collations, along with example queries for sorting characters and their corresponding outputs, to help readers determine character ordering for some of the most popular databases and collations:

MSSQL

Default collation:

```
1> SELECT CONVERT (varchar(256), SERVERPROPERTY('collation'));
2> GO
```

SQL_Latin1_General_CP1_CI_AS

(1 row affected)

```
DECLARE @input_string NVARCHAR(MAX) = '!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`
~+<=>0123456789AabBCcdDEefFGghHIijJKklLMmnNOopPQqrRSstTUuvVWwxXYyzZ

WITH SortedChars AS (
    SELECT SUBSTRING(@input_string, number, 1) AS Char
    FROM master.dbo.spt_values
    WHERE type = 'P' AND number <= LEN(@input_string)
)
SELECT STRING_AGG(Char, '') WITHIN GROUP (ORDER BY Char COLLATE SQL_Latin1_General_CP1_CI_AS) AS SortedString
FROM SortedChars;
```

Output for the above SQL.

SortedString

'!#\$%&'()*+,-./0123456789AabBCcdDEefFGghHIijJKklLMmnNOopPQqrRSstTUuvVWwxXYyzZ

MySQL and MariaDB

Default collations:

MySQL

```
mysql> SHOW COLLATION WHERE `Default` = 'Yes';
```

Collation	Charset	Id	Default	Compiled	Sortlen	Pad_attribute
armscii8_general_ci	armscii8	32	Yes	Yes	1	PAD SPACE
ascii_general_ci	ascii	11	Yes	Yes	1	PAD SPACE
big5_chinese_ci	big5	1	Yes	Yes	1	PAD SPACE
binary	binary	63	Yes	Yes	1	NO PAD
cp1250_general_ci	cp1250	26	Yes	Yes	1	PAD SPACE
cp1251_general_ci	cp1251	51	Yes	Yes	1	PAD SPACE
cp1256_general_ci	cp1256	57	Yes	Yes	1	PAD SPACE
cp1257_general_ci	cp1257	59	Yes	Yes	1	PAD SPACE
cp850_general_ci	cp850	4	Yes	Yes	1	PAD SPACE
cp852_general_ci	cp852	40	Yes	Yes	1	PAD SPACE
cp866_general_ci	cp866	36	Yes	Yes	1	PAD SPACE
cp932_japanese_ci	cp932	95	Yes	Yes	1	PAD SPACE
dec8_swedish_ci	dec8	3	Yes	Yes	1	PAD SPACE
eucjpms_japanese_ci	eucjpms	97	Yes	Yes	1	PAD SPACE
euckr_korean_ci	euckr	19	Yes	Yes	1	PAD SPACE
gb18030_chinese_ci	gb18030	248	Yes	Yes	2	PAD SPACE
gb2312_chinese_ci	gb2312	24	Yes	Yes	1	PAD SPACE
gbk_chinese_ci	gbk	28	Yes	Yes	1	PAD SPACE
geostd8_general_ci	geostd8	92	Yes	Yes	1	PAD SPACE
greek_general_ci	greek	25	Yes	Yes	1	PAD SPACE
hebrew_general_ci	hebrew	16	Yes	Yes	1	PAD SPACE
hp8_english_ci	hp8	6	Yes	Yes	1	PAD SPACE
keybcs2_general_ci	keybcs2	37	Yes	Yes	1	PAD SPACE
koi8r_general_ci	koi8r	7	Yes	Yes	1	PAD SPACE
koi8u_general_ci	koi8u	22	Yes	Yes	1	PAD SPACE
latin1_swedish_ci	latin1	8	Yes	Yes	1	PAD SPACE
latin2_general_ci	latin2	9	Yes	Yes	1	PAD SPACE
latin5_turkish_ci	latin5	30	Yes	Yes	1	PAD SPACE
latin7_general_ci	latin7	41	Yes	Yes	1	PAD SPACE
macce_general_ci	macce	38	Yes	Yes	1	PAD SPACE
macroman_general_ci	macroman	39	Yes	Yes	1	PAD SPACE
sjis_japanese_ci	sjis	13	Yes	Yes	1	PAD SPACE
swe7_swedish_ci	swe7	10	Yes	Yes	1	PAD SPACE
tis620_thai_ci	tis620	18	Yes	Yes	4	PAD SPACE
ucs2_general_ci	ucs2	35	Yes	Yes	1	PAD SPACE
ujis_japanese_ci	ujis	12	Yes	Yes	1	PAD SPACE
utf16le_general_ci	utf16le	56	Yes	Yes	1	PAD SPACE
utf16_general_ci	utf16	54	Yes	Yes	1	PAD SPACE
utf32_general_ci	utf32	60	Yes	Yes	1	PAD SPACE
utf8mb3_general_ci	utf8mb3	33	Yes	Yes	1	PAD SPACE

utf8mb4_0900_ai_ci	utf8mb4	255	Yes	Yes	0	NO PAD	
+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	+-----+	
41 rows in set (0.00 sec)							

MariaDB

```
mysql> SHOW COLLATION WHERE `Default` = 'Yes';
```

Collation	Charset	Id	Default	Compiled	Sortlen	Pad_attribute
big5_chinese_ci	big5	1	Yes	Yes	1	PAD SPACE
dec8_swedish_ci	dec8	3	Yes	Yes	1	PAD SPACE
cp850_general_ci	cp850	4	Yes	Yes	1	PAD SPACE
hp8_english_ci	hp8	6	Yes	Yes	1	PAD SPACE
koi8r_general_ci	koi8r	7	Yes	Yes	1	PAD SPACE
latin1_swedish_ci	latin1	8	Yes	Yes	1	PAD SPACE
latin2_general_ci	latin2	9	Yes	Yes	1	PAD SPACE
swe7_swedish_ci	swe7	10	Yes	Yes	1	PAD SPACE
ascii_general_ci	ascii	11	Yes	Yes	1	PAD SPACE
ujis_japanese_ci	ujis	12	Yes	Yes	1	PAD SPACE
sjis_japanese_ci	sjis	13	Yes	Yes	1	PAD SPACE
hebrew_general_ci	hebrew	16	Yes	Yes	1	PAD SPACE
tis620_thai_ci	tis620	18	Yes	Yes	4	PAD SPACE
euckr_korean_ci	euckr	19	Yes	Yes	1	PAD SPACE
koi8u_general_ci	koi8u	22	Yes	Yes	1	PAD SPACE
gb2312_chinese_ci	gb2312	24	Yes	Yes	1	PAD SPACE
greek_general_ci	greek	25	Yes	Yes	1	PAD SPACE
cp1250_general_ci	cp1250	26	Yes	Yes	1	PAD SPACE
gbk_chinese_ci	gbk	28	Yes	Yes	1	PAD SPACE
latin5_turkish_ci	latin5	30	Yes	Yes	1	PAD SPACE
armscii8_general_ci	armscii8	32	Yes	Yes	1	PAD SPACE
cp866_general_ci	cp866	36	Yes	Yes	1	PAD SPACE
keybcs2_general_ci	keybcs2	37	Yes	Yes	1	PAD SPACE
macce_general_ci	macce	38	Yes	Yes	1	PAD SPACE
macroman_general_ci	macroman	39	Yes	Yes	1	PAD SPACE
cp852_general_ci	cp852	40	Yes	Yes	1	PAD SPACE
latin7_general_ci	latin7	41	Yes	Yes	1	PAD SPACE
cp1251_general_ci	cp1251	51	Yes	Yes	1	PAD SPACE
utf16le_general_ci	utf16le	56	Yes	Yes	1	PAD SPACE
cp1256_general_ci	cp1256	57	Yes	Yes	1	PAD SPACE
cp1257_general_ci	cp1257	59	Yes	Yes	1	PAD SPACE
binary	binary	63	Yes	Yes	1	NO PAD
geostd8_general_ci	geostd8	92	Yes	Yes	1	PAD SPACE
cp932_japanese_ci	cp932	95	Yes	Yes	1	PAD SPACE
eucjpms_japanese_ci	eucjpms	97	Yes	Yes	1	PAD SPACE

```
35 rows in set (0.00 sec)
```

```
DELIMITER $$
```

```
DROP PROCEDURE IF EXISTS sort_letters$$
```

```
CREATE PROCEDURE sort_letters()
```

```
BEGIN
```

```
    DECLARE done INT DEFAULT 0;
```

```
    DECLARE word VARCHAR(256);
```

```
    DECLARE cur1 CURSOR FOR SELECT i FROM charslit;
```

```
    DECLARE CONTINUE HANDLER FOR NOT FOUND SET done = 1;
```

```
    DROP TEMPORARY TABLE IF EXISTS charslit;
```

```
    CREATE TEMPORARY TABLE charslit (i VARCHAR(256));
```

```
    INSERT INTO charslit VALUES ('!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz
```

```
    DROP TEMPORARY TABLE IF EXISTS temp;
```

```
    CREATE TEMPORARY TABLE temp (id INT, letter CHAR);
```

```
    SET @wordcount = 0;
```

```
    OPEN cur1;
```

```
    REPEAT
```

```
        FETCH cur1 INTO word;
```

```
        IF NOT done THEN
```

```
            SET @counter = 0;
```

```
            SET @len = LENGTH(word);
```

```
            WHILE (@counter <= @len) DO
```

```
                INSERT INTO temp VALUES
```

```
                (@wordcount, SUBSTRING(word, @counter, 1));
```

```
                SET @counter = @counter + 1;
```

```
            END WHILE;
```

```
            SET @wordcount = @wordcount + 1;
```

```
        END IF;
```

```
    UNTIL done END REPEAT;
```

```
    CLOSE cur1;
```

```
    SELECT GROUP_CONCAT(letter ORDER BY letter SEPARATOR '')
```

```
    FROM temp
```

```
    GROUP BY id;
```

```
END$$
```

```
DELIMITER ;
```

```
CALL sort_letters();
```

MySQL

```
mysql> SELECT @@character_set_database, @@collation_database;
```

```
+-----+-----+
| @@character_set_database | @@collation_database |
+-----+-----+
| utf8mb4                  | utf8mb4_0900_ai_ci   |
+-----+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> CALL sort_letters();
```

```
+-----+
| GROUP_CONCAT(letter ORDER BY letter SEPARATOR ")          |
+-----+
| _-,:!?. "[]{}@*/&#%`^+<=>|~$0123456789AaBbCcDdEeFfGgHhIijJkKlLmMnNoOpPqQrRsStTuUvVwWxXyYzZ |
+-----+
```

```
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

PostgreSQL

Default collation for `en-US` locale:

```
blog=# select datname,datcollate from pg_database;
```

```
datname | datcollate
```

```
-----+-----
postgres | en_US.utf8
blog      | en_US.utf8
template1 | en_US.utf8
template0 | en_US.utf8
```

```
(4 rows)
```



```

WITH t(s) AS (VALUES ('!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz
SELECT string_agg(substr(t.s, g.g, 1), "
      ORDER BY substr(t.s, g.g, 1)
    ) as char_order
FROM t
CROSS JOIN LATERAL generate_series(1, length(t.s)) g;

```

char_order

```

!#$%&'()*+,-./:;<=>?@[\]^_`{|}~$0123456789aAbBcCdDeEfGhHiIjJkKlLmMnNoOpPqQrRsStTuUvVwWxXyYzZ
(1 row)

```

SQLite

```

DROP TABLE IF EXISTS demo;
CREATE TABLE IF NOT EXISTS demo (s TEXT);

INSERT INTO demo VALUES
('!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~');

WITH cte1(r,current,rest) AS (
  SELECT rowid,substr(s,1,1), substr(s,2) FROM demo
  UNION ALL
  SELECT r,substr(rest,1,1),substr(rest,2)
  FROM cte1
  WHERE length(rest) > 0
  LIMIT 255
),
cte2 AS (SELECT * FROM cte1 ORDER BY r,current)
SELECT r,group_concat(current,"") FROM cte2 GROUP BY r;

DROP TABLE IF EXISTS demo;

```

Output for the above SQL

```

!#$%&'()*+,-./0123456789;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~

```

Semgrep Detection Rules

We have publicly published [semgrep](#) rules for detecting potentially dangerous uses of the Django, Prisma, Beego, and Entity Framework ORMs at <https://github.com/elttam/semgrep-rules> to assist

security testers and software developers.

It is important to note that these `semgrep` rules are not fully robust, due to limitations in `semgrep` itself and the inherent challenges of detecting vulnerable input parsing into ORMs.

For example, the following output from the `django-orm-dynamic-lookup-variable-key` rule did not detect the exact sink for [Label Studio CVE-2023-47117](#), but it was able to identify that significant dangerous filtering logic existed within the file.

```
$ semgrep --config /rules/python/audit/orm-leak/django-orm.yaml /app/label-studio/
```

...

```
/app/label-studio/label_studio/data_manager/managers.py
```

```
rules.python.audit.orm-leak.django-orm-dynamic-orderby-or-values
```

Dynamic field names used in `order_by()` or `values()`. Review if an attacker can influence the selected or sorted fields.

```
161 queryset = queryset.order_by(f)
```

```
rules.python.audit.orm-leak.django-orm-dynamic-lookup-variable-key <1>
```

Dynamic kwargs passed into Django ORM lookup methods with a variable key. Review if these keys can be attacker-controlled (e.g. `'user__password__contains'`). This is a primary ORM Leak sink.

```
235 filter_expressions.append(Q(**{key: int(_filter.value)}))
```

```
-----
```

```
238 filter_expressions.append(~Q(**{key: int(_filter.value)}))
```

```
-----
```

```
242 filter_expressions.append(Q(**{key + '__isnull': value}))
```

```
-----
```

```
359 q = Q(Q(**{field_name: None}) | Q(**{field_name + '__isnull': True}))
```

```
-----
```

```
359 q = Q(Q(**{field_name: None}) | Q(**{field_name + '__isnull': True}))
```

```
-----
```

```
361 q |= Q(**{field_name: ''})
```

```
-----
```

```
363 q = Q(**{field_name: [None]})
```

```
-----
```

```
366 q = Q(~Q(**{field_name: None}) & ~Q(**{field_name + '__isnull': True}))
```

```
-----
```

```
366 q = Q(~Q(**{field_name: None}) & ~Q(**{field_name + '__isnull': True}))
```

```
-----
```

```
368 q &= ~Q(**{field_name: ''})
```

```
-----
```

```
370 q = ~Q(**{field_name: [None]})
```

```
-----
```

```
390 Q(
```

```
391     **{
```

```
392         f'{field_name}__gte': _filter.value.min,
```

```
393         f'{field_name}__lte': _filter.value.max,
```

```
394     }
```

```
395 ),
```

...

<1> The `django-orm-dynamic-lookup-variable-key` rule detected dangerous Django filtering in the `apply_filter` function, which was the root cause for [Label Studio CVE-2023-47117](#).

The most effective approach we have found for identifying ORM Leak vulnerabilities is a hybrid of dynamic and static analysis: first identifying search and filtering functionality within an application dynamically, and then tracing the source code to uncover flaws in the query logic.

Closing Remarks

This will likely be our final article on the ORM Leak vulnerability class, unless we uncover a new exploitation technique targeting ORMs.

We want to emphasise that ORM Leak vulnerabilities do not depend on the use of susceptible ORMs such as Django or Prisma. Instead, this issue is widespread across modern web applications due to the growing demand for robust filtering capabilities. Developers often rely on the underlying ORM to provide sufficient validation, overlooking the fact that certain fields should never be queryable.

While SQL injection was once one of the most prevalent vulnerability classes, it is now relatively rare due to improved security awareness, training, and tooling. In contrast, we increasingly encounter applications that unintentionally allow users to search sensitive fields or control both the field and operator used in queries. As demonstrated throughout this series, these patterns can lead to significant data exposure.

We look forward to seeing further research into ORM Leak vulnerabilities and, of course, more cinematic proof-of-concept demonstrations of sensitive data being exfiltrated.

[Cruising for Shells in Flowise](#)

[Your House Has an FFmpeg Problem](#)

[Exploiting Auth0 Defaults in XSS Attacks](#)

[Jupyter Enterprise Gateway](#)

[Golang code review notes II](#)

[ORM Leaking More Than You Joined For](#)

[Gotchas in Email Parsing - Lessons From Jakarta Mail](#)

[New Method to Leverage Unsafe Reflection and Deserialisation to RCE on Rails](#)

[A Monocle on Chronicles](#)

[DUCTF 2024 ESpecially Secure Boot Writeup](#)

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